

Volume I.

Fourth Section. Symmetric functions.**§ 41. Concept of symmetric functions. Symmetric fundamental functions.**

In this section we consider whole functions, of arbitrary degree, in an arbitrary number n of variables

$$\alpha_1, \alpha_2, \dots, \alpha_n.$$

Such a function is called symmetric if it remains unchanged when the variables $\alpha_1, \alpha_2, \dots, \alpha_n$ are subjected to an arbitrary permutation; and it is such symmetric functions whose properties and laws of formation we now have to learn more precisely.

In order that a function $\Phi(\alpha_1, \alpha_2, \dots, \alpha_n)$ be symmetric, it is sufficient that it not be changed when any two of the arguments $\alpha_1, \alpha_2, \dots, \alpha_n$ are interchanged, since by § 18 all permutations can be formed by a succession of transpositions.

The function Φ will in general not be homogeneous, but will contain terms of different dimensions. If, however, one collects all terms of the same dimension, every Φ can be represented as a sum of homogeneous functions of different degrees; and, if Φ is to be symmetric, then each homogeneous constituent of a definite degree must by itself be symmetric, since the dimensions of the terms are not changed by the permutations of the variables.

Accordingly we may confine ourselves to the consideration of homogeneous symmetric functions, from which all the others can be put together.

After this it is easy to give the general form of a symmetric function. We obtain it if, in one term of such a function,

$$\alpha_1^{\nu_1} \alpha_2^{\nu_2} \dots \alpha_n^{\nu_n},$$

as in the formation of determinants, we permute the lower indices in all possible ways and take the sum of all terms so formed. Such a function may be called an elementary constituent of a symmetric function. If we take several such elementary constituents, multiply them by arbitrary factors independent of the α 's, and add them, then we obtain the most general symmetric function. The number of terms of one of these elementary constituents is $n!$ when the exponents $\nu_1, \nu_2, \dots, \nu_n$ are all different; but if one and the same exponent ν occurs several times, the permutations which give no different terms are to be omitted. For example, for three variables, if ν_1, ν_2, ν_3 are different, an elementary constituent is

$$\alpha_1^{\nu_1} \alpha_2^{\nu_2} \alpha_3^{\nu_3} + \alpha_1^{\nu_1} \alpha_2^{\nu_3} \alpha_3^{\nu_2} + \alpha_1^{\nu_2} \alpha_2^{\nu_1} \alpha_3^{\nu_3} + \alpha_1^{\nu_2} \alpha_2^{\nu_3} \alpha_3^{\nu_1} + \alpha_1^{\nu_3} \alpha_2^{\nu_1} \alpha_3^{\nu_2} + \alpha_1^{\nu_3} \alpha_2^{\nu_2} \alpha_3^{\nu_1},$$

whereas, if $\nu_2 = \nu_3$,

$$\alpha_1^{\nu_1} \alpha_2^{\nu_2} \alpha_3^{\nu_2} + \alpha_1^{\nu_2} \alpha_2^{\nu_1} \alpha_3^{\nu_2} + \alpha_1^{\nu_2} \alpha_2^{\nu_2} \alpha_3^{\nu_1}.$$

The simplest example of a symmetric function is the sum of the variables

$$\alpha_1 + \alpha_2 + \alpha_3 + \dots + \alpha_n.$$

Likewise the product

$$\alpha_1 \alpha_2 \dots \alpha_n$$

belongs here.

These two are the extreme cases of a series of symmetric functions which we call the symmetric fundamental functions and obtain as follows.

The product

$$f(x) = (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n) \quad (1)$$

is, whatever x may be, provided only that x is independent of the α 's, a symmetric function of $\alpha_1, \alpha_2, \dots, \alpha_n$. If therefore we carry out the multiplication of the several factors and arrange according to powers of x , then the coefficients of the separate powers of x are likewise symmetric functions. For they change no more under interchange of the α 's than does the function $f(x)$ (§1). We put

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n. \quad (2)$$

The functions $a_1, a_2, \dots, a_n$ are the coefficients of the algebraic equation whose roots are $\alpha_1, \alpha_2, \dots, \alpha_n$. They are called the symmetric fundamental functions. We already formed these symmetric fundamental functions in §7; as we recall, a_1 is the sum of the α 's taken with negative sign, a_2 the sum of the products two at a time, a_3 the sum of the products three at a time taken with negative sign, and so on; finally a_n is the product of all the α 's with positive or negative sign according as n is even or odd. We write, for short,

$$\begin{aligned} a_1 &= -\sum \alpha_1, \\ a_2 &= +\sum \alpha_1\alpha_2, \\ a_3 &= -\sum \alpha_1\alpha_2\alpha_3, \\ &\vdots \\ a_n &= \pm \alpha_1\alpha_2 \cdots \alpha_n. \end{aligned} \quad (3)$$

The goal of our considerations is the proof of the principal theorem that all symmetric functions of the α 's can be expressed rationally by the fundamental functions.

§ 42. The power sums.

We occupy ourselves first with another special kind of symmetric functions, the power sums. Namely, if ν denotes any positive integral exponent, then

$$s_\nu = \alpha_1^\nu + \alpha_2^\nu + \dots + \alpha_n^\nu \quad (1)$$

obviously belongs to the symmetric functions, and s_ν is called the ν th power sum. We shall derive formulae according to which the power sums are expressible by the fundamental functions.

In what follows, whenever $\varphi(x)$ is any function of x , we always denote by

$$S[\varphi(\alpha)]$$

the sum obtained by replacing x in $\varphi(x)$ successively by each of the variables $\alpha_1, \alpha_2, \dots, \alpha_n$ and adding the functions so formed, thus

$$S[\varphi(\alpha)] = \varphi(\alpha_1) + \varphi(\alpha_2) + \dots + \varphi(\alpha_n).$$

Accordingly, for example,

$$S[\alpha^\nu] = s_\nu$$

is the ν th power sum.

We now divide $f(x)$, by §4, by an arbitrary linear function $x - \alpha$ and obtain

$$\frac{f(x) - f(\alpha)}{x - \alpha} = x^{n-1} + f_1(\alpha)x^{n-2} + f_2(\alpha)x^{n-3} + \dots + f_{n-1}(\alpha), \quad (2)$$

where, by §4, (8),

$$\begin{aligned} f_1(\alpha) &= \alpha + a_1, \\ f_2(\alpha) &= \alpha^2 + a_1\alpha + a_2, \\ &\vdots \\ f_{n-1}(\alpha) &= \alpha^{n-1} + a_1\alpha^{n-2} + \dots + a_{n-1}. \end{aligned} \quad (3)$$

We put in (2), for α , each of the quantities $\alpha_1, \alpha_2, \dots, \alpha_n$, whereby $f(\alpha) = 0$, and form the sum S . The left hand side then gives, by § 14, (11),

$$nx^{n-1} + (n-1)a_1x^{n-2} + (n-2)a_2x^{n-3} + \dots + a_{n-1}, \quad (4)$$

whereas the right hand side becomes

$$nx^{n-1} + x^{n-2}S[f_1(\alpha)] + x^{n-3}S[f_2(\alpha)] + \dots + S[f_{n-1}(\alpha)]. \quad (5)$$

The comparison of coefficients of like powers of x in (4) and (5) gives

$$\begin{aligned} S[f_1(\alpha)] &= (n-1)a_1, \\ S[f_2(\alpha)] &= (n-2)a_2, \\ &\vdots \\ S[f_{n-1}(\alpha)] &= a_{n-1}. \end{aligned} \quad (6)$$

But by (1) and (3)

$$\begin{aligned} S[f_1(\alpha)] &= s_1 + na_1, \\ S[f_2(\alpha)] &= s_2 + a_1s_1 + na_2, \\ &\vdots \\ S[f_{n-1}(\alpha)] &= s_{n-1} + a_1s_{n-2} + a_2s_{n-3} + \dots + na_{n-1}. \end{aligned}$$

Consequently from (6) one obtains the following system of formulae due to Newton:

$$\begin{aligned} 0 &= s_1 + a_1, \\ 0 &= s_2 + a_1s_1 + 2a_2, \\ 0 &= s_3 + a_1s_2 + a_2s_1 + 3a_3, \\ &\vdots \\ 0 &= s_{n-1} + a_1s_{n-2} + a_2s_{n-3} + \dots + (n-1)a_{n-1}. \end{aligned} \quad (7)$$

This system of formulae can, however, be continued still further; for, since

$$S[f(\alpha)] = 0, \quad S[\alpha f(\alpha)] = 0, \quad S[\alpha^2 f(\alpha)] = 0, \dots,$$

it follows that

$$\begin{aligned} 0 &= s_n + a_1s_{n-1} + a_2s_{n-2} + \dots + a_n, \\ 0 &= s_{n+1} + a_1s_n + a_2s_{n-1} + \dots + a_ns_1, \\ 0 &= s_{n+2} + a_1s_{n+1} + a_2s_n + \dots + a_ns_2, \quad \dots \end{aligned} \quad (8)$$

By the formulae (7), (8) the problem is now solved of representing, one after another, the functions $s_1, s_2, \dots$ up to any height as whole rational functions of the symmetric fundamental functions; for example

$$\begin{aligned} s_1 &= -a_1, \\ s_2 &= a_1^2 - 2a_2, \\ s_3 &= -a_1^3 + 3a_1a_2 - 3a_3, \\ s_4 &= a_1^4 - 4a_1^2a_2 + 4a_1a_3 + 2a_2^2 - 4a_4. \end{aligned} \quad (9)$$

The manner of formation of these expressions shows that the coefficients in these representations are whole numbers.

Conversely, by means of the formulae (7) and (8), one can express the symmetric fundamental functions $a_1, a_2, a_3, \dots$ rationally by the power sums $s_1, s_2, s_3, \dots$. This representation is, however, far less simple in so far as the coefficients are not whole but fractional numbers, for example

$$a_1 = -s_1, \quad 2a_2 = s_1^2 - s_2, \quad 6a_3 = -s_1^3 + 3s_1s_2 - 2s_3. \quad (10)$$

One can also continue the system of formulae (8) in the opposite direction, if one forms the equations

$$S[\alpha^{-1}f(\alpha)] = 0, \quad S[\alpha^{-2}f(\alpha)] = 0, \dots$$

In this way one obtains a means of expressing the power sums s_ν also for negative exponents ν by the fundamental functions. These sums of negative powers likewise belong to the symmetric functions, although no longer to the whole, but to the fractional, ones. They pass into whole functions only after multiplication by powers of the product $\alpha_1\alpha_2 \cdots \alpha_n$. For completeness we put down here the first two of these formulae:

$$\begin{aligned} 0 &= a_{n-1} + a_n s_{-1}, \\ 0 &= 2a_{n-2} + a_{n-1}s_{-1} + a_n s_{-2}. \end{aligned} \quad (11)$$

§ 43. Proof of the principal theorem for two variables.

We now pass to the proof of the fundamental theorem of the theory of symmetric functions, that they are all rationally expressible by the symmetric fundamental functions. Since we use complete induction as the means of proof, we first derive the theorem under the assumption that only two independent variables α, β are given, but in a way which at the same time will bring out the guiding thought for the general proof.

We denote the symmetric fundamental functions by

$$a = -(\alpha + \beta), \quad b = \alpha\beta \quad (1)$$

and accordingly put

$$f(x) = (x - \alpha)(x - \beta) = x^2 + ax + b. \quad (2)$$

Let $S(\alpha, \beta)$ be any whole rational and symmetric function of α and β . We can substitute for β from (1) the value $-(\alpha + a)$, and obtain, if we arrange according to powers of α ,

$$S(\alpha, \beta) = S(\alpha, -\alpha - a) = A_0\alpha^m + A_1\alpha^{m-1} + \cdots + A_m, \quad (3)$$

where the coefficients $A_0, A_1, \dots, A_m$ depend only on a and on the coefficients still occurring in S . We note expressly, however, that if in $S(\alpha, \beta)$ no fractional numerical coefficients occur, then no fractions occur in the coefficients $A_0, A_1, \dots, A_m$ either.

We now put

$$\Phi(x) = A_0x^m + A_1x^{m-1} + \cdots + A_{m-1}x + A_m \quad (4)$$

and divide $\Phi(x)$ by $f(x)$, by § 3. We obtain a quotient Q and a remainder which, with respect to x , is at most of the first degree; thus

$$\Phi(x) = Qf(x) + A + Bx. \quad (5)$$

Here A and B are now whole functions of a and b , and they contain no fractional numerical coefficients if none occur in S . If now we put $x = \alpha$, it follows from (5) and (3), since $f(\alpha)$ vanishes, that

$$S(\alpha, \beta) = A + B\alpha. \quad (6)$$

But since $S(\alpha, \beta)$ and likewise A, B are symmetric, it follows by interchanging α and β that

$$S(\alpha, \beta) = A + B\beta. \quad (7)$$

From this one concludes, since α and β are independent variables, that $B = 0$, and consequently

$$S(\alpha, \beta) = A, \quad (8)$$

which proves the fundamental theorem in this case.